

1 Methodology: The Minimal Decoupling Model (MDM)

We define the economy as a dynamical system governed by the interplay between capital accumulation, technological diffusion, and labor participation. The system state vector is defined as $x = [K_H, K_L, R, \theta, P, S]$, representing High-Tech Capital, Low-Tech Capital, Automation Capital, Diffusion, Labor Participation, and the Accumulated Subsidy, respectively.

1.1 Production and Labor Demand

Output is generated via two sectors. The High-Tech sector (Y_H) utilizes a Constant Elasticity of Substitution (CES) production function, allowing for substitutability between cognitive labor (L_H) and automation (R):

$$Y_H = A_H K_H^\alpha [(1 - \mu)L_H^\rho + \mu(\psi(\theta)R)^\rho]^{\frac{1}{\rho}}$$

where $\rho = (\sigma - 1)/\sigma$ determines the elasticity of substitution σ , and $\psi(\theta)$ represents the efficiency advantage of automation, which decays as technology diffuses ($\theta \rightarrow 1$). The Low-Tech sector (Y_L) follows a Cobb-Douglas specification:

$$Y_L = A_L K_L^\beta L_L^{1-\beta}$$

Labor is allocated to equate the Marginal Product of Labor (MPL) across sectors, defining the market wage w :

$$w = \frac{\partial Y_H}{\partial L_H} = \frac{\partial Y_L}{\partial L_L}$$

1.2 Participation Dynamics and the Social Contract

Standard models assume inelastic labor supply. We model Participation (P) as a dynamic state driven by the gap between the effective wage (w_{eff}) and a reservation "survival" wage (w_{bar}). We employ a hyperbolic tangent function to model the saturation of the labor force and the friction of social change:

$$\dot{P} = \begin{cases} \eta P \tanh(\lambda(w_{\text{eff}} - w_{\text{bar}})) & \text{if } w_{\text{eff}} \geq w_{\text{bar}} \\ \delta_P P \tanh(\lambda(w_{\text{eff}} - w_{\text{bar}})) & \text{if } w_{\text{eff}} < w_{\text{bar}} \end{cases} - \chi(\mu\psi)^\gamma(1 - \theta)P$$

The final term represents "Skill Exclusion," where rapid technological shifts disenfranchise a fraction of the workforce regardless of wage levels.

1.3 Closed-Loop Control: The Thermostat

To maintain participation at a target level (P_{target}), we introduce a Proportional-Integral (PI) Controller acting on the effective wage. The control input, the wage subsidy fraction $S(t)$, evolves to minimize the error $e(t) = P_{\text{target}} - P(t)$:

$$S(t) = \text{sat}_{[0, S_{\text{max}}]} \left(K_p e(t) + K_i \int_0^t e(\tau) d\tau \right)$$

The effective wage perceived by the workforce is thus augmented by the subsidy, funded by a fraction of total output Y :

$$w_{\text{eff}} = w_{\text{market}} + \frac{S(t) \cdot Y}{P \cdot N}$$

1.4 Investment Friction

To reflect realistic constraints on capital scaling, we modify the standard accumulation equation $\dot{K} = I - \delta K$ by introducing an adjustment cost factor ϕ . The effective capital formation is given by:

$$\dot{K}_i = \frac{I_i}{1 + \phi \left(\frac{I_i}{K_i} \right)} - \delta_i K_i \quad \text{for } i \in \{H, L, R\}$$

As investment intensity (I/K) rises, the marginal efficiency of investment declines, preventing instantaneous singularities in output growth.